Comprehensive Road Scene Understanding for Autonomous Driving: Semantic Segmentation to Anomaly Detection with EoMT

Omer Ozkan - Ceylin Afyoncu - Franck Hernold Kewo Kameni
Politecnico di Torino
Student IDs: s360761, s346382, s359890
{s360761, s346382, s359890}@studenti.polito.it

Abstract

Semantic and anomaly segmentation is critical for autonomous driving. But predefined-class models fail on the unknown objects, and it creates danger in the real world scenarios. We evaluate EoMT (encoder-only mask transformer with DINOv2 backbone) on semantic segmentation, and we compare COCO-trained EoMT (55.17% mIoU) with Cityscapes-trained EoMT (81.68% mIoU) on Cityscapes validation dataset. We fine-tune the COCO-EoMT on Cityscapes which reaches to 78.85% mIoU. For anomaly segmentation, we compare ERFNet and EoMT models using four post-hoc scoring methods which are MSP, MaxLogit, MaxEntropy, and RbA on 5 different datasets. Our findings show that EoMT-Cityscapes with MSP reaches 74.36% AuPRC on RoadAnomaly21. Our code is available at https://github.com/omerzkan/eomt-anomaly-segmentation.git.

1. Introduction

Semantic segmentation analyzes every pixel and classifies each one into related class. A vehicle needs to know where pedestrians are and where the road is. Without pixel-level analysis, this task is difficult. The Cityscapes dataset [6] is a standard benchmark for this type of task. It includes 5000 photos of German streets. Each photo is manually labeled with 19 classes such as road, sidewalk, building. ERFNet [13] is a traditional pixel-based approach. It uses a CNN (convolutional neural network) with an encoder-decoder architecture, and classifies each pixel independently. MaskFormer [4] changed this approach by predicting N binary masks and N class labels, then combining them. In this way, the model handles both semantic and panoptic segmentation [9]. Mask2Former [5] improved MaskFormer by adding masked cross-attention, so the model only looks at the image region where each mask is predicted. EoMT [8] simplified this further. It uses only the DINOv2 [12] encoder and removes the pixel decoder entirely. The problem is that all these models are closed-set. They were trained on exactly 19 classes, and during training they did not see anything except what were trained on. However, the model must assign objects it has never seen to one of its known classes. if the model assigns an animal or fence as road, the vehicle tries to drive through it. This creates safety problems. This type of data is called out-of-distribution (OoD) data. We use four post-hoc scoring methods (MSP, MaxLogit [7], MaxEntropy [3], RbA [11]) with temperature scaling, and evaluate them on benchmarks from SMIYC [2] and Fishyscapes [1]. We evaluated EoMT on semantic segmentation and compared two differently trained models (COCO-EoMT vs Cityscapes-EoMT). Then we fine-tuned the COCO-EoMT on Cityscapes, and the model achieved 78.85% mIoU.

2. Methodology

2.1. Semantic Segmentation

Table 1. Semantic segmentation results on `Cityscapes val` (mIoU %). EoMT-COCO is evaluated zero-shot with COCO→Cityscapes class mapping. EoMT-Fine-tuned starts from EoMT-COCO and is fine-tuned on Cityscapes.

Model	mIoU
ERFNet	72.50
EoMT-COCO (zero-shot)	55.17
EoMT-Cityscapes	**81.68**
EoMT-Fine-tuned	78.85

Semantic segmentation was our first task, and we used EoMT [8], which is a mask-based architecture built on a DINOv2-pretrained Vision Transformer [12]. We compared two EoMT models using two different checkpoints: EoMT-COCO and EoMT-Cityscapes. EoMT-COCO is trained on the COCO dataset [10], a large general-purpose dataset with 118k training images covering daily scenes such as streets, kitchens, beaches, and animals. COCO has 133 class labels: 80 “thing” classes such as person, car, dog, and 53 “stuff” classes such as sky, grass, wall. This makes the model strong on a wide range of visual objects, but it is not road-specific. The second model is EoMT-Cityscapes, trained on the Cityscapes dataset [6]. Cityscapes is a smaller dataset (around 3k training images) of road and driving scenes from German cities. It has 19 class labels specifically chosen for autonomous driving, such as road,traffic sign, rider. It is road-specific but not as broad as COCO. We evaluated both models on the Cityscapes validation set, which contains 500 images. The goal is to understand how much a general model loses compared to a specialized one, and whether fine-tuning close the gap. We need to translate the COCO predictions into the Cityscapes label space before evaluation. We built a `coco_to_cityscapes` mapping that works in three steps:

- **Direct name match:** obvious matches found by string comparison. For example, COCO “car” maps directly to Cityscapes “car”, and COCO “person” to Cityscapes “person”.
- **Synonyms:** some classes are named differently in the two datasets. We built a synonym dictionary to handle these cases. For example, COCO includes sidewalks in “pavement-merged”.
- **Ignored:** COCO labels with no Cityscapes equivalent are mapped to the ignore index 255, which excludes them from evaluation. For example, if EoMT-COCO predicts “dog”, that pixel is ignored.

Intersection-over-Union (IoU) is a method to measure the accuracy for each class:

$$\mathrm{IoU}_c = \frac{TP_c}{TP_c + FP_c + FN_c} \tag{1}$$

Mean IoU (mIoU) is the average of these scores.

2.2. Fine-Tuning

We wanted to see if we could adapt it to the Cityscapes domain without losing its general knowledge. For fine tuning drop the classification head because COCO’s classification head produces 133 logits per query. We replaced it with a new head that has 19 outputs which is one per Cityscapes class. The rest of the weights are kept frozen. For efficiency, we froze the DINOv2 backbone and trained the queries, and the new classification head. We also use Automatic Mixed Precision (AMP). Instead of FP32, we use mixed-precision FP16. For the training setup 20 epochs, $K = 200$ queries, image size 640×640 used. Training was done on a single NVIDIA L4 GPU via Google Colab Pro. For the loss function we used standard Mask2Former [5] loss. We use AdamW with a learning rate of 1×10^{-4}.

2.3. Anomaly Segmentation

We evaluate ERFNet and EoMT under anomaly segmentation. Anomaly segmentation detects pixels belonging to objects which are not found in the training class set, out-of-distribution. For evaluation, we use post-hoc methods which use class logits, and calculate anomaly score per pixel. Both ERFNet (pixel-based) and EoMT (mask-based) produce class logits, and this allows us to use the same scoring functions for both models.

Post-hoc anomaly detection methods.

- **MSP (Maximum Softmax Probability):** uses the highest softmax probability as an uncertainty indicator.

$$\mathrm{score}_{\mathrm{MSP}}(\mathbf{x}) = 1 - \max_c p_c(\mathbf{x}) \tag{2}$$

- **MaxLogit** [7]: uses the raw logits before softmax, because they preserve the magnitude of the model’s confidence.

$$\mathrm{score}_{\mathrm{MaxLogit}}(\mathbf{x}) = -\max_c L_c(\mathbf{x}) \tag{3}$$

- **MaxEntropy** [3]: if the probability is spread across many classes (high entropy), the model is uncertain and the pixel is a possible anomaly. If it puts all probability on one class (low entropy), the model is confident.

$$\mathrm{score}_{\mathrm{Entropy}}(\mathbf{x}) = -\sum_{c=1}^{C} p_c(\mathbf{x}) \log p_c(\mathbf{x}) \tag{4}$$

- **RbA (Reweighted by Area) [11]:** Only for mask-based models. RbA flags pixels that are not strongly claimed by any mask. It relies on an independence assumption. In a mask architecture, each class behaves like a one-vs-all classifier, so class probabilities do not need to sum to 1. In a pixel-wise classifier, the class probabilities must sum to 1, it violates the independence assumption, and RbA loses the OoD-detection meaning. That’s why we don’t apply RbA to ERFNet.

$$\mathrm{score}_{\mathrm{RbA}}(\mathbf{x}) = -\sum_{c=1}^{C} \tanh(L_c(\mathbf{x})) \tag{5}$$

Temperature scaling. It is a calibration technique that can shift recall and precision in anomaly detection. It divides logits by T before softmax. $T > 1$ makes probabilities more uncertain, $T < 1$ makes them more confident. We tested $T \in \{0.5, 0.75, 1.0, 1.1, 1.5, 2.0\}$ on MSP, MaxEntropy, and RbA. We didn’t test on MaxLogit, because it doesn’t use softmax.

Datasets and metrics. As evaluation benchmarks, we use a smaller subsets of these five datasets which are provided by course instructors. These are:

- **RoadAnomaly21 (from SMIYC [2]):** 10 images, large unknown objects in diverse environments.
- **RoadObstacle21 (from SMIYC [2]):** 30 images, small unknown objects on the road.
- **Fishyscapes Lost&Found [1]:** 100 validation images of real lost-cargo objects.
- **Fishyscapes Static [1]:** 30 images, synthetic anomalies on the road scenes.
- **RoadAnomaly (from SMIYC [2]):** 60 images, earlier version of SMIYC's anomaly track.

We use two metrics:

- **AuPRC** (Area under the Precision-Recall Curve): Evaluates pixel-level separability. Higher values mean better, used for discrimination of rare anomalies.
- **FPR@95** (False Positive Rate at 95% True Positive Rate): Measures false positive rate when the true positive rate is fixed at 95%. It is a safety critical metric. Lower is better.

3. Results

3.1. Semantic Segmentation

EoMT-Cityscapes is the best model with 81.68% mIoU among the compared model. Beside them ERFNet (72.50%) is competitive for a pixel-based model.

Table 2. Selected per-class IoU (%) on `Cityscapes val`. EoMT-COCO scores 0% on classes with no COCO equivalent.

Class	EoMT-COCO	EoMT-Cityscapes
pole	**0.00**	71.04
rider	**0.00**	71.11
traffic sign	**4.74**	82.13
road	94.13	98.40
car	84.33	95.54
bicycle	73.34	81.27
Mean	**55.17**	**81.68**

EoMT-COCO scores 0% IoU on `pole, rider`, and near-0% on `traffic sign` (Table 2) because COCO has no equivalent label. It is not the problem of the model but a label space mismatch. It is the motivation for fine-tuning. Fine-tuning lifts mIoU from 55.17 to 78.85, recovers the most of the gap. This result shows that COCO's DINOv2-backed visual priors work well, the only problem was the classification head, not the features.

3.2. Anomaly Segmentation

Pixel-based vs mask-based. We see these results from the comparison of ERFNet and EoMT (Table 3):

- On RoadAnomaly-21 dataset, ERFNet's MaxLogit is 38.31, and EoMT-Cityscapes' MaxEntropy is 77.88. There are 40 point difference
- On RoadObstacle-21 dataset, ERFNet's MaxLogit is 4.63 and EoMT-Cityscapes' MSP is 90.11. There are 85% difference
- On RoadAnomaly dataset, ERFNet's MaxLogit 15.58 and EoMT-Cityscapes' MaxEntropy 78.14. There are 63% difference

With these results, we find that on most benchmarks, the EoMT's models substantially outperform the ERFNet. Because mask-based classification creates smooth shapes because it looks at the whole object at once. However, pixel-softmax looks at each pixel one by one, that's why it is vulnerable to noise.

Effect of fine-tuning. To see the effect of the fine-tuning, we compare the EoMT-Cityscapes with EoMT-Fine-tuned on AuPRC (Table 3).

- On RoadAnomaly-21 dataset: MSP score drops from 74.36% to 57.15% (17% change)
- On RoadObstacle-21 dataset: MSP score drops from 90.11% to 73.61% (16% change)
- On RoadAnomaly dataset: MSP score drops from 75.69% to 54.99% (21% change)
- On FS L&F dataset: MSP score increases from 25.42% to 27.11% (2% change)

According to these results, EoMT-COCO recovers closed set mIoU (Table 3) but decrease the anomaly scores compared to training from scratch on Cityscapes. This happens because training the model for a short time makes it hyperfocused on normal objects, and this ruins its ability to spot unusual things.

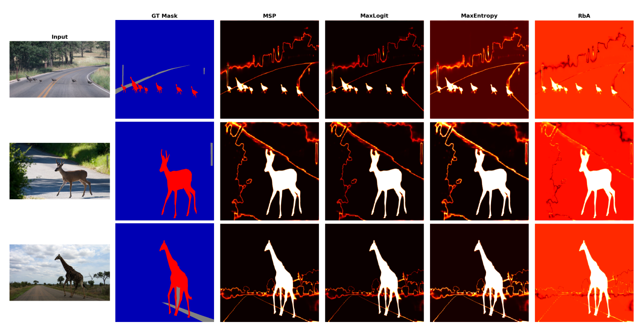

Figure 1. Qualitative comparison of anomaly score maps on RoadAnomaly21. Each row shows an input image alongside score maps produced by MSP, MaxLogit, MaxEntropy, and RbA for EoMT-Cityscapes.

Comparison of scoring functions. Interesting finding here is that MaxLogit beats MSP and MaxEntropy on ERFNet (38.31 vs 29.09 vs 30.96 on RA-21), but on EoMT they are almost same. This result shows that scoring-method choice depends on the architecture.

Table 3. Temperature scaling results across the three models. Only $T \in \{0.5, 1.0, 2.0\}$ shown; intermediate values $T \in \{0.75, 1.1, 1.5\}$ follow the same trend. AuPRC (↑, %) and FPR95 (↓, %) on five benchmarks. RbA is omitted for ERFNet as it is incompatible with pixel-based architectures.

			RA-21		RO-21		FS L&F		FS Static		RoadAnom	
Model	Method	T	AuPRC↑	FPR95↓	AuPRC↑	FPR95↓	AuPRC↑	FPR95↓	AuPRC↑	FPR95↓	AuPRC↑	FPR95↓
ERFNet	MSP	0.5	27.05	62.75	2.42	63.24	1.28	66.73	6.60	43.48	12.19	82.04
		1.0	29.09	62.57	2.71	65.27	1.75	50.62	7.47	41.84	12.42	82.59
		2.0	30.60	65.05	3.00	72.74	2.69	47.92	9.54	41.29	12.64	84.58
	MaxEntropy	0.5	28.58	62.74	2.64	63.73	1.59	52.82	7.04	43.20	12.33	82.26
		1.0	30.96	62.67	3.04	65.95	2.58	50.19	8.84	41.55	12.67	82.76
		2.0	30.27	67.09	2.97	75.98	3.51	47.10	11.72	41.75	12.79	85.60
EoMT-Cityscapes	MSP	0.5	74.21	33.73	90.10	0.55	25.40	14.91	54.33	34.94	75.51	19.37
		1.0	74.36	33.73	90.11	0.55	25.42	14.90	54.35	34.94	75.69	19.31
		2.0	74.55	33.73	90.11	0.55	25.44	14.90	54.36	34.93	75.87	19.28
	MaxEntropy	0.5	77.43	33.68	90.09	0.59	29.55	14.83	53.21	34.91	78.43	19.03
		1.0	77.88	33.60	90.01	0.68	29.56	14.72	52.51	34.84	78.14	18.79
		2.0	77.91	33.53	89.97	0.72	29.01	14.65	51.46	34.77	77.43	18.63
	RbA	0.5	69.57	76.02	73.77	99.95	26.00	67.27	55.85	84.55	74.12	44.02
		1.0	69.99	80.06	86.97	99.95	25.97	8.79	56.11	82.68	74.62	20.41
		2.0	70.83	91.98	88.25	99.94	25.92	10.05	55.91	35.88	75.09	17.96
EoMT-Fine-tuned	MSP	0.5	58.59	98.60	73.69	25.10	27.07	26.22	53.35	39.78	54.96	16.27
		1.0	57.15	98.61	73.61	25.09	27.11	26.16	53.41	39.74	54.99	16.22
		2.0	56.61	98.62	73.57	25.09	27.13	26.15	53.43	39.72	55.00	16.20
	MaxEntropy	0.5	54.10	98.71	73.43	28.83	27.91	26.21	53.87	39.53	55.83	15.28
		1.0	54.49	98.82	72.62	57.49	28.56	26.33	54.27	39.25	56.22	14.78
		2.0	54.65	98.82	71.69	60.82	29.18	26.24	54.50	38.76	56.33	14.51
	RbA	0.5	51.52	99.53	62.10	99.98	31.42	78.97	56.29	89.51	52.39	81.57
		1.0	51.82	99.85	64.22	99.98	30.61	26.92	56.29	90.34	56.14	13.38
		2.0	51.88	99.87	66.12	99.98	30.36	24.49	55.73	89.65	56.57	12.09

RbA has a different pattern in here. It achieves highest AuPRC on FS L&F (30.61) and FS Static (56.29) for EoMT-Fine-tuned. However, it also produces high score for FPR95, so it separates the outlier but it is too sensitive so it makes a lot of false alarms.

Qualitative results. Figure 1 shows anomaly score maps for the four scoring methods on representative Road-Anomaly21 images.

3.3. Temperature Scaling Study

For temperature scaling, we tried $T \in \{0.5, 0.75, 1.0, 1.1, 1.5, 2.0\}$ on MSP, MaxEntropy, and RbA for EoMT-Cityscapes, EoMT-Fine-tuned, and ERFNet. Main finding that AuPRC is mostly stable across T (Table 3).

- In EoMT-Cityscapes (Table 3), MSP AuPRC on RA-21 stays in a 0.3% window, and MaxEntropy stays in a 0.5% window.
- In EoMT-Fine-tuned (Table 3), It follows similar pattern as EoMT-Cityscapes
- In ERFNet (Table 3), MSP on RA-21 has a change range for 3%, but its AuPRC is so low that the gain is not meaningful.

Interesting finding that RbA's AuPRC is stable, but its FPR95 swings with T:

- EoMT-Cityscapes RbA on FS L&F: FPR95=67.27 at T=0.5 vs 8.79 at T=1.0
- EoMT-Cityscapes RbA on RoadAnomaly: FPR95's score decreases from 20.41 at T=1.0 to 17.96 at T=2.0.
- EoMT-Fine-tuned RbA on RoadAnomaly: FPR95 = 81.57 at T=0.5 vs 12.09 at T=2.0

As result, we found that T=1.0 is competitive for AuPRC most settings. Temperature scaling doesn't meaningfully improve AuPRC, and can shift the FPR95 results, especially for RbA, but it is not a consistent gain. Post-hoc temperature calibration is not a reliable method for improving anomaly segmentation on these benchmarks.

4. Discussion

Mask-based architectures help anomaly detection. Results show us that EoMT performs much better than ERFNet on anomaly detection. On RA-21, EoMT achieves 74.36 AuPRC when ERFNet is only 38.31. Similar Things are observed on RO-21 (90.11 vs 4.63) and RoadAnomaly (78.14 vs 15.58). Possible reason for this result is that EoMT uses object queries. Each query behaves like a classifier. On the other hand, pixel-wise classifiers assign each pixel to a known class, and it makes anomaly detection harder.

Fine-tuning recovers most of the closed-set gap. Fine tuning substantially increases the performance of EoMT-COCO. The mIoU increases from 55.17 to 78.85. It almost closes the performance gap with EoMT-Cityscapes which has a mIoU score of 81.68. In the fine-tuning, the biggest advantage is that only the segmentation head and object queries were trained, while the backbone left frozen. For training, 20 epochs with NVIDIA L4 GPU used. However, even if it increase the mIoU score, at the same time it reduces the anomaly detection performance. For example, the MSP score on RA-21 drops from 74.36 to 57.15. This showed us a trade-off between semantic segmentation performance and anomaly detection capability.

Limitations. In this project we had several limitations. The main limitation was limited computational sources. We only used Colab environment for evaluation and training, which allows limited GPU power, only 20 training epochs and no backbone fine-tuning. Since RbA achieves strong AuPRC, this observation indicates room for improvement.

Future work. Several improvements can be done in future work. One option is to unfreeze the backbone layers to improve accuracy. Also, evaluating the approaches on a complete benchmarks would provide more comprehensive performance and results.

5. Conclusion

In this work, we studied semantic segmentation and anomaly segmentation for road-scene by comparing EoMT and ERFNet models. We fine-tuned EoMT-COCO on Cityscapes training dataset using frozen-backbone, and evaluated using four anomaly scoring functions on five benchmarks. Results showed us that fine-tuning recovers most of the performance gap between COCO-pretrained model and the native Cityscapes model. Finally, we saw that EoMT consistently outperforms ERFNet in anomaly detection. Overall, these results suggest that mask classification is a better option for safety-critical systems that require both semantic understanding and anomaly detection.

Acknowledgments

- We declare that Generative AI tools were not used for writing the project report
- For code development of the project, we used Claude Opus 4.7 and 4.6 by Anthropic. This tool is used for these reasons:
  - Debugging errors, catching bugs, fixing memory issues during large-scale evaluation
  - Understanding the EoMT and ERFNet code structure
  - Reviewing the formulas, and help with the structuring the repository
  - Performance optimization, helping to change data-loading patterns that reduced runtime

All generated code was reviewed, tested, and integrated by the authors. All decisions like model choice, training configuration, evaluation protocol, dataset selection and analysis of experimental results were made by the authors.

References

[1] Hermann Blum, Paul-Edouard Sarlin, Juan Nieto, Roland Siegwart, and Cesar Cadena. Fishyscapes: A benchmark for safe semantic segmentation in autonomous driving. In *ICCV Workshops*, 2019. 1, 3

[2] Robin Chan, Krzysztof Lis, Svenja Uhlemeyer, et al. Segmentmeifyoucan: A benchmark for anomaly segmentation. In *NeurIPS Datasets and Benchmarks*, 2021. 1, 3

[3] Robin Chan, Matthias Rottmann, and Hanno Gottschalk. Entropy maximization and meta classification for out-of-distribution detection in semantic segmentation. In *ICCV*, 2021. 1, 2

[4] Bowen Cheng, Alexander G Schwing, and Alexander Kirillov. Per-pixel classification is not all you need for semantic segmentation. In *NeurIPS*, 2021. 1

[5] Bowen Cheng, Ishan Misra, Alexander G Schwing, Alexander Kirillov, and Rohit Girdhar. Masked-attention mask transformer for universal image segmentation. In *CVPR*, 2022. 1, 2

[6] Marius Cordts, Mohamed Omran, Sebastian Ramos, et al. The cityscapes dataset for semantic urban scene understanding. In *CVPR*, 2016. 1, 2

[7] Dan Hendrycks, Steven Basart, Mantas Mazeika, et al. Scaling out-of-distribution detection for real-world settings. In *ICML*, 2022. 1, 2

[8] Tom Kerssies and Daan de Geus. An encoder-only approach to mask classification. In *CVPR*, 2025. 1

[9] Alexander Kirillov, Kaiming He, Ross Girshick, et al. Panoptic segmentation. In *CVPR*, 2019. 1

[10] Tsung-Yi Lin, Michael Maire, Serge Belongie, et al. Microsoft coco: Common objects in context. In *ECCV*, 2014. 1

[11] Nishant Nayal, Yigit Yavuz, João F Henriques, and Daan de Geus. Scaling out-of-distribution detection for real-world settings. In *ECCV*, 2024. 1, 2

[12] Maxime Oquab, Timothée Darcet, Théo Moutakanni, et al. Dinov2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024. 1

[13] Eduardo Romera, José M Alvarez, Luis M Bergasa, and Roberto Arroyo. Erfnet: Efficient residual factorized convnet for real-time semantic segmentation. pages 263–272, 2018. 1